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THE PALEOMAGNETISM OF CRETACEOUS ROCKS FROM ISRAEL

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The original directions of magnetization of oriented samples from two extensive lava fields in Israel, one of Lower and one of Upper Cretaceous age, show considerable scatter ($K \simeq 2$) that is reduced appreciably upon AC magnetic cleaning ($K \simeq 50$). The resulting pole positions are: 105.8E, 59.5N for Lower Cretaceous and 95.8E, 41.64N for Upper Cretaceous.

Since azimuthal errors in the field orientations were suspected, a statistical model is proposed in which the declination and inclination have independent Gaussian distributions. Correlation coefficients regressions, and confidence intervals computed from the above data show that the proposed statistical model is reasonable for one volcanic unit, but does not seem appreciably different from Fisherian statistics for the second. The statistical results indicate a marked change in confidence limits by use of the bivariate model thus indicating the inadequacy of the Fisher statistics for some paleomagnetic purposes.

1. Introduction

Erosinal remanents of several extensive Cretaceous lava fields occur in Israel [1]. In southern Israel, in the Ramon anticline region (see fig. 1), this lava sequence is both underlain and overlain by Lower Cretaceous sandstones. In northern Israel the lavas are interbedded in a Marine sequence of Upper Cretaceous (Senonian) age and are confined to outcrops in the Carmel region [2, 3]. All of the lavas have undergone some deformation involving tilting and folding. Nevertheless, the structures are open and structural corrections can readily be made. Field sampling was done with a portable gasoline powered core drill [4] and all orientations were made with a magnetic compass.

Twenty-two field cores were drilled from three outcrop areas of the Lower Cretaceous lavas in the Ramon anticline. The lavas in this area show no evidence of extensive alteration and apparently have undergone little or no weathering. Most of the samples [11] come from two sites on a single conical hill which rises above the surrounding area.

Fifteen cores were obtained from one structural block of the Upper Cretaceous lavas in the Carmel area of northern Israel. These lavas have been extensively altered, perhaps, by weathering during Cretaceous time. This alteration has effected the natural remanent magnetization (NRM) and is reflected in the initial scatter of the directions of NRM.

2. Laboratory measurements

Directions of magnetization were measured by means of a spinner magnetometer operating at 271 Hz. All samples were demagnetized using 420 Hz alternating current demagnetization equipment in steps of

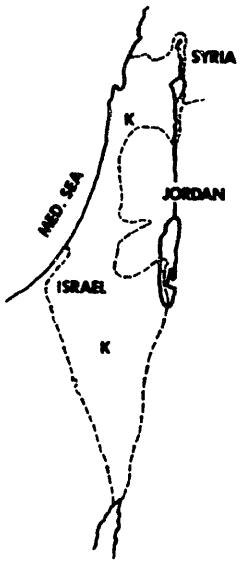


Fig. 1. Map showing location of sampling localities (K); northern locality is Upper Cretaceous, southern locality is Lower Cretaceous.

peak-field values of 44, 88, 175, 350, 700 Oe, and in some cases up to 2800 Oe.

The results of this demagnetization are shown in fig. 2 for the Lower Cretaceous samples, and fig. 3 for the Upper Cretaceous samples. Figs. 2(a) and 3(a) show the directions of magnetization, before any cleaning or structural correction. Figs. 2(b) and 3(b) show the changes in direction of magnetization of selected samples upon demagnetization, and figs. 2(c) and 3(c) show the final directions of all samples after AC demagnetization and structural correction.

The directions of magnetization for the Upper Cretaceous samples converged very well upon AC demagnetization and were considered to have been successfully demagnetized when the angular changes occuring in the last 3 successive demagnetization steps were less than 5 degrees.

The directions of magnetization observed in Lower Cretaceous samples did not converge as well upon AC demagnetization even when magnetic fields of up to 2800 Oe were applied. Several of the samples that did not converge reversed their direction of motion during the step by step demagnetization.

In order to ascertain the effect of incomplete removal of secondary components a best fitting great circle was calculated for each set of demagnetization results for each sample. This procedure allows one to project the demagnetization of the sample beyond the last data point if necessary, provided that the NRM consists of only two components. If the samples are sufficiently scattered, as are the ones from the Lower Cretaceous outcrops (fig. 2), the intersection of all of the resulting great circles should provide an independent determination of the mean demagnetized direction.

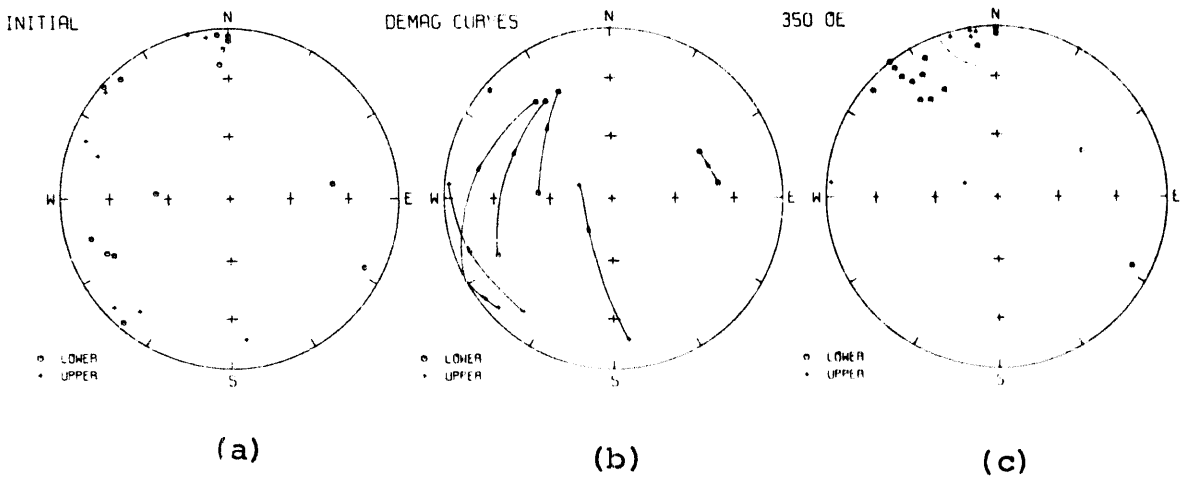


Fig. 2. NRM and demagnetization results from the Lower Cretaceous of Israel. Those samples separated by the dashed line in the right hand figure are all from site I.

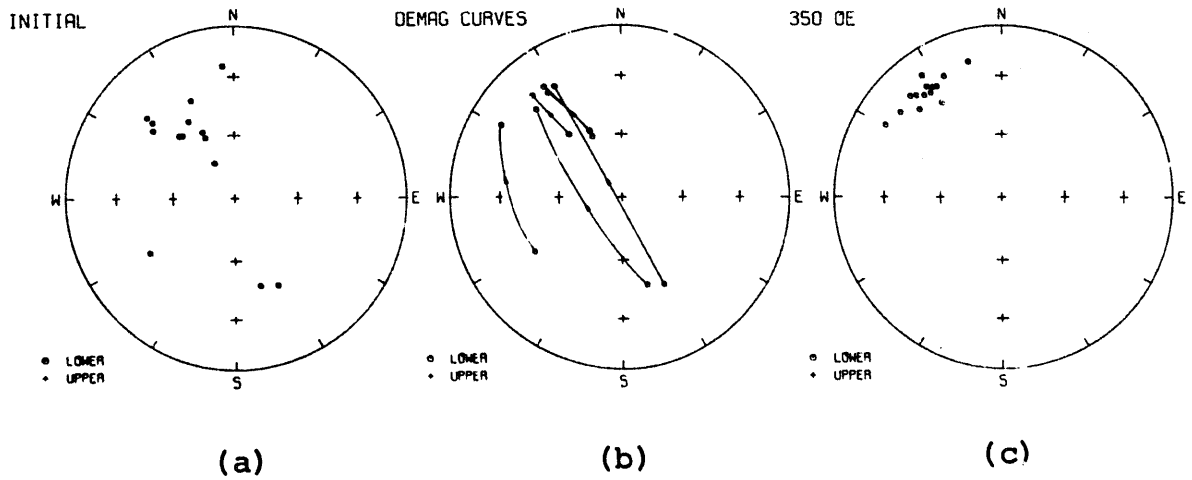


Fig. 3. NRM and demagnetization results from the Upper Cretaceous lavas exposed in the Carmel region of northern Israel.

Great circle computations have been made for each sample from two of the three collecting localities. The directions of magnetization for the third locality (site I) did not change appreciably on demagnetization and thus great circle routes could not be calculated. The results of the great circle analyses are shown in fig. 4 and listed in table 1 along with the mean declinations and inclinations from conventional statistical analysis [5].

Most of the samples from site III reversed their direction of movement on progressive demagnetization above fields of 175 Oe. Thus, two great circle solutions have been made for these samples, one for the prograde portion of the demagnetization path and one for the retrograde portion of the path. These are shown by the letters P and R respectively, in fig. 4.

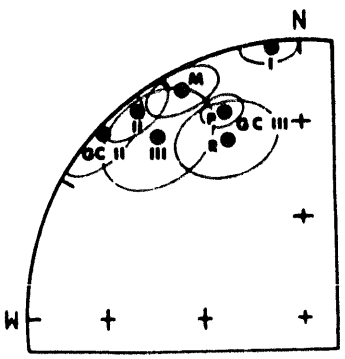


Fig. 4. Site means for the Lower Cretaceous sample sites I, II, and III. M is the mean of all sites. Also shown are the results of great circle analyses (GC II and GC III). The great circle analysis of site III has been separated into prograde (P) and retrograde (R) determinations.

The direction of magnetization derived from the great circle intersections from site II agrees rather well with the mean direction of the demagnetized NRM data. However, the results from site III do not agree as well. The NRM data for site III was initially as scattered as that for site II but did not group as well upon demagnetization primarily because most samples changed their direction of movement during the progressive demagnetization. The prograde and retrograde great circle analyses agree very well and thus suggest that the direction of magnetization corresponding to minimum scatter during demagnetization may be considerably in error, up to 15°, primarily in declination. The disagreement between the minimum scatter point and the great circle intersections probably is due to the demagnetization process having begun to remove the higher coercive force component responsible for the retrograde motion prior to complete removal of the softer component responsible for prograde motion.

The broken line in fig. 2(c) separates one of the Lower Cretaceous outcrops (site I) showing a systematic difference in direction of magnetization. Moreover, the samples from this site are initially well grouped and show almost no change on AC demagnetization. This outcrop may have undergone

Table 1
Magnetization directions and pole positions for Cretaceous rocks from Israel.

	Dec.	Inc.	*N*	*R*	*K*	alpha	delta	Long.	Lat.	*dp*	*dm*
Lower Cretaceous											
NRM of all samples	355.6	5.3	22	14.86	2.9	22.1	46.2	135.8	61.7	11.1	22.2
Demag. Loc. 1 350 Oe	353.2	0.8	7	6.94	100.3	6.1	7.5	131.8	59.1	3.0	6.0
Demag.* Loc. 2	321.6	5.5	7	6.89	52.8	8.4	10.3	85.0	44.2	4.2	8.4
Demag.* Loc. 3	321.3	18.3	7	6.66	17.6	14.8	17.9	77.2	48.2	8.0	15.4
Demag.* Loc. 2 and 3	321.4	11.8	14	13.46	24.2	8.3	15.9	81.4	46.2	4.3	8.4
Demag. all at 350 Oe	327.1	4.0	22	19.20	7.5	12.1	28.9	92.1	48.2	6.9	13.5
Demag.* (one omitted)	332.3	8.4	21	19.63	14.6	8.6	20.7	94.9	52.8	4.4	8.7
Gt. circle II	312.8	−0.2	7			10.8		80.4	35.7		
Gt. circle III P	339.5	22.4	7			4.5		95.7	63.1		
Gt. circle III R	337.9	31.0	7			13.4		84.7	65.5		
Upper Cretaceous											
NRM of all samples	327.2	34.6	15	12.45	5.5	18.0	33.4	71.5	57.7	11.9	20.0
Demag. all at 350 Oe	325.5	−6.2	15	14.73	51.5	5.4	10.9	95.8	41.6	2.7	5.4

* Peak demagnetizing fields vary between 175 Oe and 1400 Oe.

a structural rotation different from that of the other two localities, but if so, this change was not noticed in the field. Alternatively, the local present magnetic field may have been different at this outcrop and therefore produced a systematic declination error since the samples were oriented by means of a magnetic compass. Because of this uncertainty, mean directions and pole positions have been computed both with and without incorporation of this locality and are summarized in table 1. One sample has been omitted from the analysis summarized in table 1 since it did not converge upon demagnetization and is separated from the other demagnetized samples by more than 30°.

3. Statistical treatment of results

Scatter in the direction of magnetization in samples magnetized at the same time may be due to: (1) the physical process of magnetization (for which we assume small errrors normally distributed about the mean direction); (2) highly localized secondary processes (lightning, weathering, structural deformation, etc.); (3) sampling error – particularly when azimuth is determined by a magnetic compass; or (4) laboratory measurement errors.

Errors due to (1) and (4) are generally negligible, and errors involving (2) can be removed in most cases by applying structural corrections and magnetic cleaning. Sampling errors (3) generally are small, however, where orientation is done by magnetic compass, potentially large errors in orientation due to local distortion of the earth's field may be present.

When structural corrections are not necessary, orientation errors in inclination rarely exceed two degrees provided that the sampling is done carefully. Consequently, the errors in inclination will result primarily from the random small errors expected from (1), (2) and (4) while the errors in declination will be due to the potentially large errors arising from (3) as well as from random small errors from (1), (2) and (4). The errors in declination and inclination, therefore, can be virtually independent in certain situations.

Since all field orientations were determined by a magnetic compass and the structural corrections were small ($< 15°$), the following bivariate statistical analysis is suggested as being more representative of the true dispersion of the sample population than would be the normal Fisher analysis [5].

4. Estimation of means

The bivariate normal density function is discussed in most statistical texts [6, p. 303] and is defined as

$$f(D,I) = \frac{1}{2\pi S_D S_I \sqrt{1-\rho^2}} \exp\left[\frac{-1}{2(1-\rho^2)}\right.$$

$$\left.\times\left\{\left(\frac{D-\bar{D}}{S_D}\right)^2 - 2\rho\left(\frac{D-\bar{D}}{S_D}\right)\left(\frac{I-\bar{I}}{S_I}\right)+\left(\frac{I-\bar{I}}{S_I}\right)^2\right\}\right]$$

where $\bar{D}$ and $\bar{I}$ are the means of the declination D and inclination I, S_D and S_I the standard deviations and ρ the correlation coefficient. This equation is not strictly valid for analysis of directions on a sphere because for D and I do not have distributions between + and − infinity. However, if the population under analysis has a relatively small scatter, say less than ± 45° in both D and I, and has inclination values distributed about 0, then the difference between the speherical distribution in D and I and the x-y distribution on the plane tangent to the sphere at the mean direction of the population is relatively small. Under these conditions it is approximately valid to treat the data as a bivariate distribution in x and y if one remembers that the confidence limits thus derived will not exactly match those of Fisher [5] because the spherical and planar coordinate systems diverge as one departs from the point of tangency.

From the discussions given above we can assume that D and I are virtually independent and thus can be treated as variables of two independent univariate normal density functions. This is equivalent to assuming that $\rho = 0$ in the above equation.

The confidence intervals for the means thus can be shown to be given by

$$\left[\bar{D} - \frac{S_D}{\sqrt{n}} t_{(\alpha/2,\, n-1)}\right] < \bar{D} < \left[\bar{D} + \frac{S_D}{\sqrt{n}} t_{(\alpha/2,\, n-1)}\right]$$

$$\left[\bar{I} - \frac{S_I}{\sqrt{n}} t_{(\alpha/2,\, n-1)}\right] < \bar{I} < \left[\bar{I} + \frac{S_I}{\sqrt{n}} t_{(\alpha/2,\, n-1)}\right]$$

where $1-\alpha$ is the degree of confidence, n is the number of samples, and t is taken from tables for $1-\alpha = 0.95$.

5. Test of statistical assumptions

Inasmuch as we initially assumed D and I were independent (i.e. $\rho = 0$) we should test this assumption by first computing an estimate of the correlation coefficient (ref. [6], p. 310)

$$\hat{\rho} = \frac{\sum_i [I_i - \bar{I}]\,[D_i - \bar{D}]}{\sqrt{\sum_i (I_i - \bar{I})^2}\sqrt{(D_i - \bar{D})^2}}$$

and then testing the hypothesis that $\rho = 0$ against the value of $\hat{\rho}$. This can be done by means of the statistic (ref. [6], p. 310)

$$Z_{\alpha/2} = \frac{\sqrt{n-3}}{2} \ln \frac{(1+\hat{\rho})(1-\rho)}{(1-\hat{\rho})(1+\rho)}.$$

If $Z_{\alpha/2}$ is less than 1.96, the hypothesis that $\rho = 0$ is valid at the confidence level $1-\alpha = 0.95$.

6. Numerical results

The numerical evaluation of the above relations when applied to the Lower Cretaceous and Upper Cretaceous data are given in table 2. The correlation between D and I for both the Upper Cretaceous and the Lower Cretaceous samples is small and therefore the bivariate statistical model is applicable. The slightly more significant correlation between D and I for the Lower Cretaceous samples (using the data from all samples, $n = 22$ in table 2) implies that this set of data may have some dependence between D and I. A possible reason for this correlation is a small relative error in structural correction between the three collecting sites. Since each site has a different declination (see fig. 2), small errors in the amount of structural correction (in effect errors in inclination) might produce an apparent dependence between D and I and thus result in an increased correlation when a bivariate analysis is made on these samples.

An alternate reason for the apparent correlation between D and I is that one sample is very far removed from the rest of the samples (40° in D and 60° in I). Since the demagnetization behavior of this sample indicated that a large portion of the secondary magnetization was not removed even at fields of 1400 Oe, it is probably justifiable to remove

Table 2

	Upper Cretaceous	Lower Cretaceous	
n	15	22	21
$\bar{D}$	325.5	326.4	332.1
$\bar{I}$	−6.1	2.8	8.0
S_D	10.6	22.8	18.2
S_I	4.2	20.9	11.5
$\hat{\rho}$	0.238	0.293	−0.275
$Z_{0.025}$	0.843	1.32	−1.20
Confidence intervals	$319.6 < D < 331.4$ $-8.4 < I < -3.8$	$316.2 < D < 336.6$ $-6.5 < I < 12.1$	$328.8 < D < 340.4$ $2.8 < I < 13.3$

it from the statistical calculations. The calculation for $n = 21$ in table 2 has eliminated this divergent sample and considerable improvement is indicated by the change in S_D and S_I. The correlation between D and I remains small and changed its slope.

Thus one can conclude that the bivariate model in which D and I are independent is valid for both the Upper and Lower Cretaceous suites. Since S_D and S_I are now equal, the use of the bivariate analysis results in a more meaningful estimate of the amount of uncertainty present in the observation.

7. Comparison with Fisher statistics

The bivariate distribution approaches to a good approximation the Fisher distribution [5] when S_D and S_I are equal. The standard Fisher analysis gives the results shown in table 1. The mean declination, inclinations and confidence limits for the Fisher analyses of the Lower and Upper Cretaceous results are compared with those of the bivariant analysis in table 3.

Table 3

	n	D	I	α_D^*	α_I^*
Upper Cretaceous					
Bivariate	15	325.5	−6.1	5.9	2.3
Fisher	15	325.5	−6.2	5.4	5.4
Lower Cretaceous					
Bivariate	21	332.1	8.0	8.3	5.2
Fisher	21	332.3	8.4	8.6	8.6

* α_D and α_I for the bivariate analysis are the confidence intervals for D and I respectively and are α_{95} for the Fisher analysis.

The declinations and inclinations are very similar as would be expected. The standard deviations (see tables 1 and 2) and confidence intervals, however, are quite different with those from the Fisher analysis always being greater than either of the standard deviations of the bivariant analysis. This is probably due to the fact that the use of a bivariate distribution is not strictly valid as was discussed above. The most marked difference is between the confidence intervals for the inclination of the Upper Cretaceous samples. The decrease in the standard deviation by a factor of 2.7 when the bivariant analysis is used points out the usefulness of this type of analysis for certain types of paleomagnetic data.

8. Pole positions

Assuming a dipole field at the time of magnetization of the samples and using Fisher statistics, the average pole positions, and their respective principal values of the ellipse of confidence are as is shown in table 4.

The confidence limits for the bivariate analysis of the Upper Cretaceous data are compared to the Fisher confidence oval in fig. 5. Relative to the estimate of the Fisher analysis the estimate of the mean inclination for the Upper Cretaceous sample is

Table 4
The average pole positions and their respective principal values of the ellipse of confidence.

			Fisher		Bivariate	
	Longitude	Latitude	*dp*	*dm*	*dp*	*dm*
Upper Cretaceous	95.8W	41.6N	2.1	5.4	1.5	6.0
Lower Cretaceous	105.8W	59.5N	6.32	12.5	2.5	8.2

Table 5
Summary of Israel, Africa and Middle East poles.

Africa [7]							
Lupata Volcanics			7 sites	339.0	3.0	100.5W	62.0N
Africa [10]							
Mlanje Massif	333.0	−54.0	1 site	36.0	9.5	108.0W	60.0N
Turkey [9]							
Upper Cretaceous	345.0	40.0	6 sites	90.0	7.0	98.5W	69.5N
Upper (?) Cretaceous	153.5	−36.5	6 sites	35.0	11.0	81.5W	60.5N
Cretaceous	140.0	−36.0	2 sites	8.0	–	64.0W	51.0N
Syria [8]							
Aptian	313.5	9.5	2 sites	–	5.5	78.0W	38.0N
Israel (this paper)							
Upper Cretaceous	325.5	−6.2	2 sites	51.5	5.4	95.8W	41.6N
Lower Cretaceous	332.3	8.4	3 sites	14.6	8.6	94.9W	52.8N

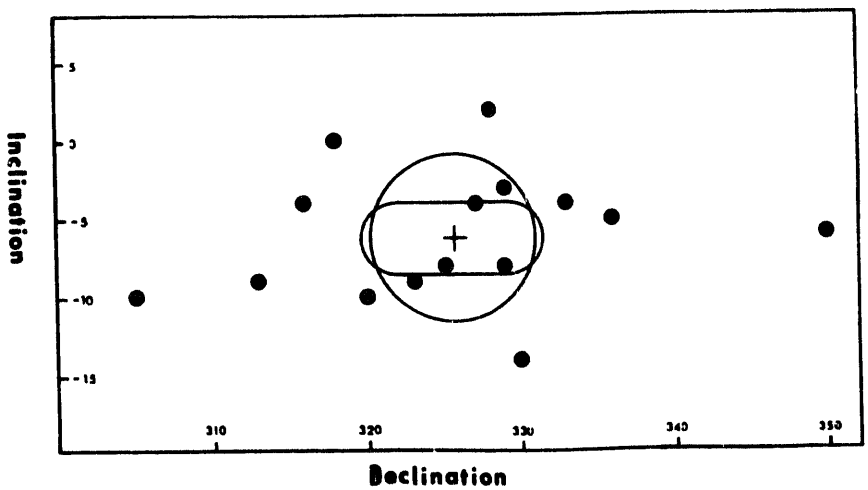


Fig. 5. Comparison of confidence limits derived from bivariate and Fisher statistics.

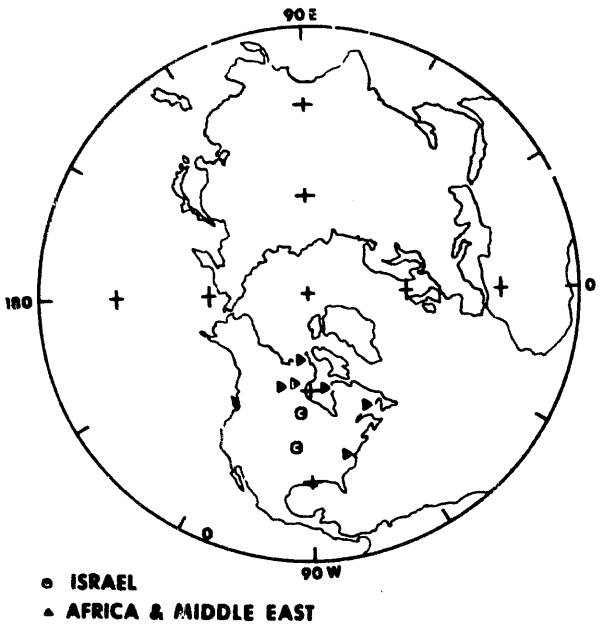


Fig. 6. Comparison of pole positions for Israel with those from Africa and the Middle East.

improved significantly, while the estimate of the declination remains about the same. Similar relations hold for the Lower Cretaceous except the differences are not as marked. Thus the distance to the pole is known with a much higher precision than is the azimuth to the pole.

The Cretaceous paleomagnetic results from Turkey, Africa, and Syria are summarized in table 5 and are compared with the results from Israel in fig. 6. The average pole position reported by Gough and Opdyke [7] for the Lower Cretaceous Lupata Volcanics from Southern Africa lies 0.5 within the confidence region of the Lower Cretaceous results from Israel. The paleomagnetic results from rocks of Aptian age from Lebanon and Syria [8] are quite close to the results obtained from the Upper Cretaceous Volcanics in Israel.

The similarity in directions of magnetization in the Middle East and Southern Africa gives support to the idea that the continental block south of the Alpine fold belt has behaved as one structural block at least since Lower Cretaceous time. More detailed sampling may possibly reveal the tectonic history of this African-Arabian block, and thus could be used in reconstructing and understanding of the appearance of the Syrian-African rift zone, extending from Lebanon to southern East Africa.

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